Fiscal Year:	FY 2025	**Task Last Updated:**	FY 10/10/2024
PI Name:	Martin, Bryn Ph.D.		
Project Title:	Ophthalmic and Intracranial Structural Changes in Head-Down Tilt Bedrest: Potential Countermeasures and Comparison to SANS Findings in Astronauts (OPTICS study)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	bmartin@neelatx.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	330-475-9747
Organization Name:	Neela Therapeutics		
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City:	Moscow	**State:**	ID
Zip Code:	83843	**Congressional District:**	1
Comments:	PI moved to Alcyone Therapeutics, Inc., in late 2020 from University of Idaho (Ed., December 2021) PI moved to Neela Therapeutics in late 2025 from Alcyone (Ed, November 2025)		
Project Type:	Ground	**Solicitation / Funding Source:**	2018-2019 HERO 80JSC018N0001-SANS: Spaceflight Associated Neuro-ocular Syndrome Countermeasures. Appendix C
Start Date:	12/08/2021	**End Date:**	12/07/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/07/2025 per NSSC information (Ed., 11/14/24). NOTE: End date of new grant 80NSSC22K0023 is 12/07/2024 per grant information documents (Ed., December 2021). NOTE: End date of previous grant number 80NSSC20K0920 changed to 10/12/2020 per JSC Grants Admin/NSSC information, due to PI move and new grant established (Ed., 12/10/21).		
Key Personnel Changes/Previous PI:	There have not been any key personnel changes in the last reporting period.		
COI Name (Institution):	Kramer, Larry M.D. (University of Texas Health Science Center at Houston) Laurie, Steven Ph.D. (KBR/NASA Johnson Space Center) Macias, Brandon Ph.D. (NASA Johnson Space Center) Marshall-Goebel, Karina Ph.D. (KBR/NASA Johnson Space Center) Williams, Michael M.D. (University of Washington, Seattle) Loerch, Linda Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	80NSSC22K0023		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	NOTE: Continuation of "Ophthalmic and Intracranial Structural Changes in Head-Down Tilt Bedrest: Potential Countermeasures and Comparison to SANS Findings in Astronauts (OPTICS study)," grant 80NSSC20K0920, with the same Principal Investigator (PI) Dr. Bryn Martin. The proposed research plan builds on our group's ongoing research to quantify ophthalmic and intracranial changes in long duration spaceflight astronauts using novel non-invasive magnetic resonance imaging (MRI) protocols and post-processing techniques. We propose to use identical techniques currently implemented pre- and post-spaceflight to also quantify how long-duration bed rest and countermeasures applied during bed rest impact ophthalmic and intracranial structures. It is critically essential to use identical techniques that we have already applied in astronauts to determine if structural alterations seen in bed rest are similar to what our team has already quantified in astronauts with spaceflight associated neuro-ocular syndrome (SANS). Similarly, it is essential to use the identical techniques to quantify the capability of a countermeasure to mitigate structural alterations in bed rest.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will help better understand ophthalmic and cerebrospinal fluid physiology and anatomy that can potentially be used for diagnostic and therapeutic applications of terrestrial human diseases such as hydrocephalus and cerebrospinal fluid drug delivery.
Task Progress:	Spaceflight has been hypothesized to result in a bulk body fluid shift toward the head. The OPTICS study is conducted on a group of healthy subjects undergoing a head-down tilt (HDT) procedure in an attempt to mimic fluid shifts observed in space, and implement potential countermeasures to evaluate their potential efficacy in mitigating the negative effects of long-duration spaceflight on sensitive eye and brain tissues due to fluid shifts. This research builds on our group’s ongoing research to quantify ophthalmic and intracranial changes in long-duration spaceflight in astronauts, using novel non-invasive magnetic resonance imaging (MRI) protocols and post-processing techniques. Key progress in the 2024 fiscal year involved data analyses for all 48 participants enrolled in the study to determine parameters related to ophthalmic and intracranial structural and physiological changes in head-down tilt bedrest and the potential impact of countermeasures. We also continued quantification of the same parameters in long-duration spaceflight astronauts to determine how similar they are impacted compared to head-down tilt bedrest. In brief, the parameters related to five domains: 1) optic nerve and globe structural alterations; 2) cerebrospinal fluid flow; 3) intracranial blood flow; 4) intracranial tissue and fluid volumetrics; 5) intracranial fluid and tissue shifts. Additionally, the following novel MRI-based parameters were developed in 2024: a) anterior globe shift (proptosis or exophthalmos); b) brain tissue shift within the skull; c) cerebrospinal fluid shift within the skull; d) spinal subarachnoid space volume distribution, total spinal volume and length, and spinal cord tortuosity. Data analyses are ongoing and pending group comparisons. In fiscal year 2025, we plan to complete and publish the key findings related to understanding the potential impact of countermeasures, comparison of groups with astronauts and controls, and exploring hypotheses further evaluating brain shift, cerebrospinal fluid shift, and optic globe proptosis in astronauts.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Abstracts for Journals and Proceedings	Williams M, Warthen K, Sater S, Kramer L, Hasan K, Macias B, Laurie S, Martin BA. "Quantification of brain and CSF 3D center of mass shift in astronauts and ground-based controls." Hydrocephalus 2024, the 16th Meeting of the Hydrocephalus Society, Nagoya, Japan, September 13-16, 2024. Abstracts. Hydrocephalus 2024, the 16th Meeting of the Hydrocephalus Society, Nagoya, Japan, September 13-16, 2024. , Sep-2024
Abstracts for Journals and Proceedings	Sater SH, Warthen KG, Bangudu O, Kramer LA, Hasan KM, Williams MA, Macias BR, Laurie SS, Martin BA. "Semi-automated MRI-based segmentation of the spinal central nervous system for quantification of spinal changes in head-down-tilt subjects." 2024 NASA Human Research Program Investigators’ Workshop, Galveston, TX, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators’ Workshop, Galveston, TX, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Sater SH, Warthen KG, Kramer LA, Hasan KM, Williams MA, Macias BR, Laurie SS, Martin BA. "Posterior optic globe volume displacement during 30-day head-down tilt bedrest and the effect of potential countermeasures." 2024 NASA Human Research Program Investigators’ Workshop, Galveston, TX, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators’ Workshop, Galveston, TX, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Warthen KG, Sater SH, Kramer LA, Hasan KM, Williams MA, Macias BR, Laurie SS, Martin BA. "Long-duration spaceflight alters intracranial tissue and fluid position." 2024 NASA Human Research Program Investigators’ Workshop, Galveston, TX, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators’ Workshop, Galveston, TX, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Martin BA, Warthen KG, Sater SH, Kramer LA, Hasan KM, Williams MA, Macias BR, Laurie SS. "Ophthalmic and intracranial structural changes in head-down tilt bed rest (OPTICS): Update on findings and comparison to astronauts." 2024 NASA Human Research Program Investigators’ Workshop, Galveston, TX, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators’ Workshop, Galveston, TX, February 13-16, 2024. , Feb-2024

Articles in Peer-reviewed Journals	Carter KJ, Laurie SS, Warthen KG, Sater SH, Martin BA, Feng CM, Young M, Hasan KM, Kramer LA, Macias BR. "Normal variation in brain volumetrics, CSF dynamics, and ocular structures from magnetic resonance images of healthy participants over two years." J Appl Physiol (1985). 2025 Jul 1;139(1):1-13. https://doi.org/10.1152/japplphysiol.00596.2024 ; PMID: 40397516 , Jul-2025